Université Saint-Joseph de Beyrouth
Faculté d'ingénierie

SnowFLOWS: Generative Spatio–Temporal Modelling of Snow Cover Dynamics Using Satellite Imagery

MDP26–19

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ESIB Multidisciplinary Project
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Introduction

Spatio–Temporal Snow Forecasting using Deep Learning and Satellite Data

Project Team & Multidisciplinary Context

- Computer & Communications Engineering (CCE) students
- Collaboration between data processing, modeling, and system design

Project Context

- Snow cover plays a critical role in water resources in Lebanon
- Monitoring snow dynamics is essential for environmental and economic planning

Problem Statement

- Difficulty in accurately forecasting snow evolution over time
- Limited temporal coverage (Sentinel–2) vs limited spatial resolution (MODIS)
- High variability of snow behavior in mountainous regions

Engineering Challenge

- Design a system capable of modeling complex spatio–temporal snow dynamics
- Integrate multi–source satellite data
- Capture nonlinear temporal dependencies using machine learning

Project Objectives and Outcomes

Project Objectives

- Develop a snow forecasting system using satellite data
- Model temporal dynamics of snow cover using LSTM networks
- Analyze the impact of sequence length on prediction performance

Project Outcomes

- Built a multi–source dataset (Sentinel–2, MODIS, meteorological data)
- Achieved improved performance over baseline models
- Identified trade–off between numerical accuracy and visual realism

Satellite Data → Preprocessing → LSTM Model → Forecast

Why ? Because …

- Snow behavior is nonlinear and time–dependent
- Traditional methods fail to capture complex temporal patterns
- Deep learning enables data–driven modeling of environmental systems

Project Functional Requirements

Data Requirements

- ✓ Extract satellite data (Sentinel–2, MODIS)
- ✓ Ensure temporal consistency and alignment
- ✓ Build structured time–series dataset

Model Requirements

- ✓ Predict weekly snow cover
- ✓ Capture temporal dependencies
- ✓ Support sequence–based input (LSTM)

Feature Requirements

- ✓ Integrate multi–source variables (snow, temperature, precipitation)
- ✓ Encode seasonality (cyclical features)
- ✓ Normalize data for stable training

Evaluation Requirements

- ✓ Compare with baseline persistence model
- ✓ Use MAE and RMSE
- ✓ Analyze different sequence lengths

Project Constraints

1. Data Constraints

- Cloud coverage affecting Sentinel–2 observations
- Limited temporal coverage of high–resolution Sentinel–2 data

2. Technical Constraints

- Trade–off:

	Spatial	Temporal
MODIS	Low ↓	High ↑
Sentinel–2	High ↑	Low ↓

- Processing limitations for large multi–year satellite datasets

3. Model Constraints

- Sensitivity to sequence length selection
- Risk of overfitting with long sequences
- Difficulty capturing extreme snow peaks accurately

Standards Applied
• ESA Copernicus Sentinel–2 data format compliance • NASA MODIS data usage standards • NDSI–based snow detection (remote sensing standard) • Standard ML pipeline: normalization + train/test split

Related Works

Approach	Strengths	Limitations	Suitability for Snow Forecasting
ARIMA	Simple, interpretable	Linear assumptions	↓ Poor for nonlinear snow dynamics
Random Forest	Handles nonlinear data	No temporal memory	! Limited for time series
LSTM (our choice)	Captures temporal dependencies	Needs tuning	✓ **Best suited**

Key Insight

- Snow forecasting requires modeling nonlinear temporal behavior
- LSTM is more suitable for spatio–temporal satellite data

Solution Design

Overview

Data Extraction → Preprocessing → Sequence Construction → LSTM Model → Prediction → Evaluation & Visualization

Goal of the system

- End–to–end framework to forecast weekly snow cover using satellite data and machine learning.

Data Sources

- Sentinel–2 (ESA): high–resolution optical imagery (10m) for detailed snow detection
- MODIS (NASA): long–term medium–resolution data (~500m) for temporal consistency
- ERA5: meteorological reanalysis (temperature & precipitation)

Core Idea

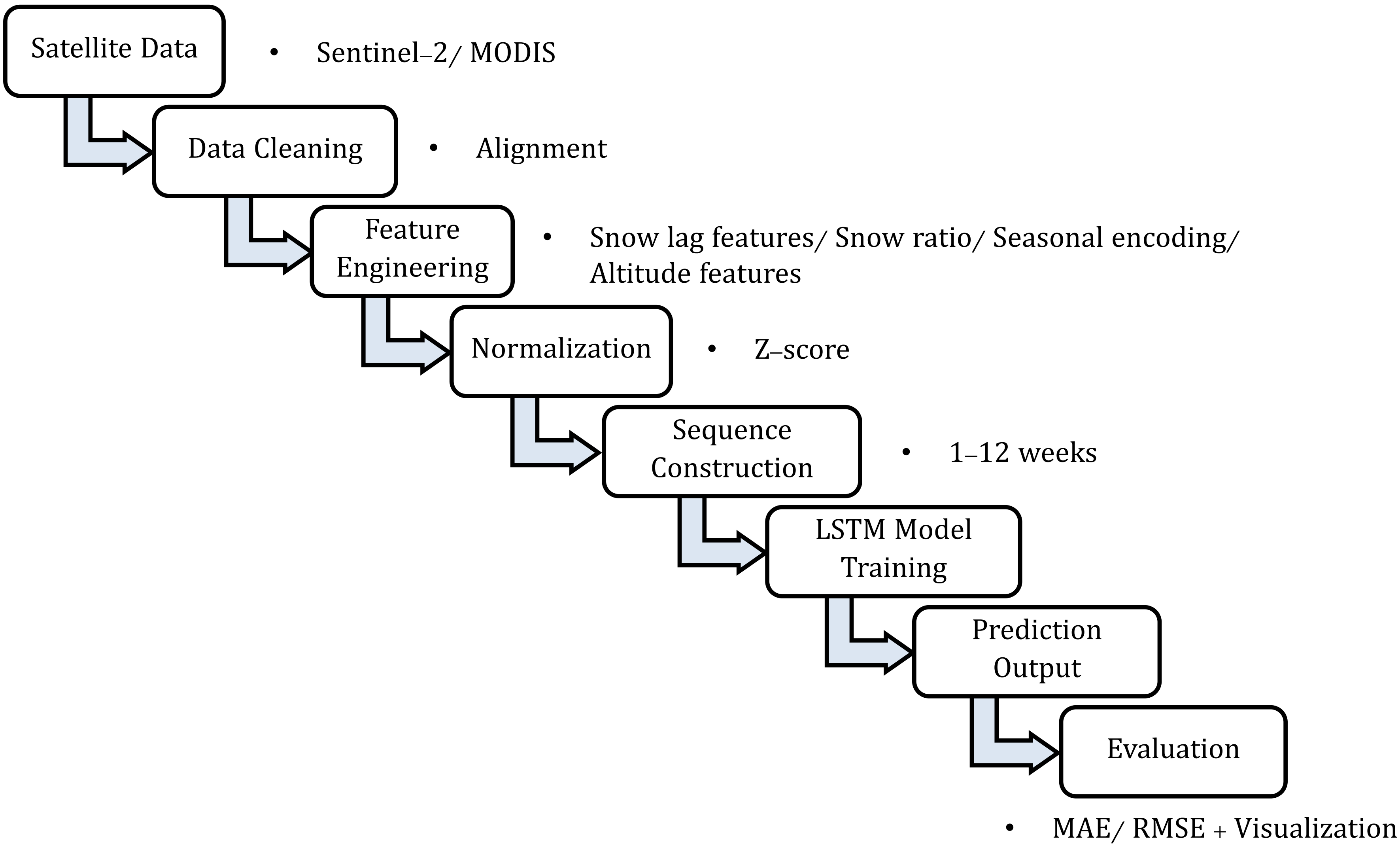
- Transform raw satellite observations into structured time–series inputs for deep learning–based forecasting.

Model Choice

- LSTM selected due to its ability to model:
  - temporal dependencies
  - nonlinear environmental dynamics

Solution Design

Conceptual Model of SnowFLOWS System



Project Management

- We have 9 main phases:

	A	B	C	D	E	F	G	H	I	J	K
1	**Phase**	**Task Description**	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Week 9**
2	**Phase 1**	Data Acquisition & Exploration									
3	**Phase 2**	Data Preparation (Cleaning, Normalization)									
4	**Phase 3**	Dataset Construction & Feature Engineering									
5	**Phase 4**	Sequence Construction									
6	**Phase 5**	Model Development (LSTM Setup)									
7	**Phase 6**	Training & Initial Evaluation									
8	**Phase 7**	Iterative Improvements (Sequences + Tuning)									
9	**Phase 8**	Extended Experiment (MODIS 25 Years)									
10	**Phase 9**	Final Analysis & Report Writing									

Iterative Workflow

- Continuous refinement of:
  - dataset structure
  - sequence length
  - model architecture
- Multiple experiments run per configuration
- Decisions based on averaged results (MAE/RMSE)

Conflict Management

Model instability across runs → solved using multiple executions + averaging

Sequence length trade-off:

- 6 weeks → best numerical results
- 8 weeks → best visual realism

→ both analyzed and justified

Data alignment issues (Sentinel vs MODIS) → resolved via temporal synchronization

Solution Implementation

Study Area & Data Sources

Study area: Lebanon (FAO/GAUL boundaries)

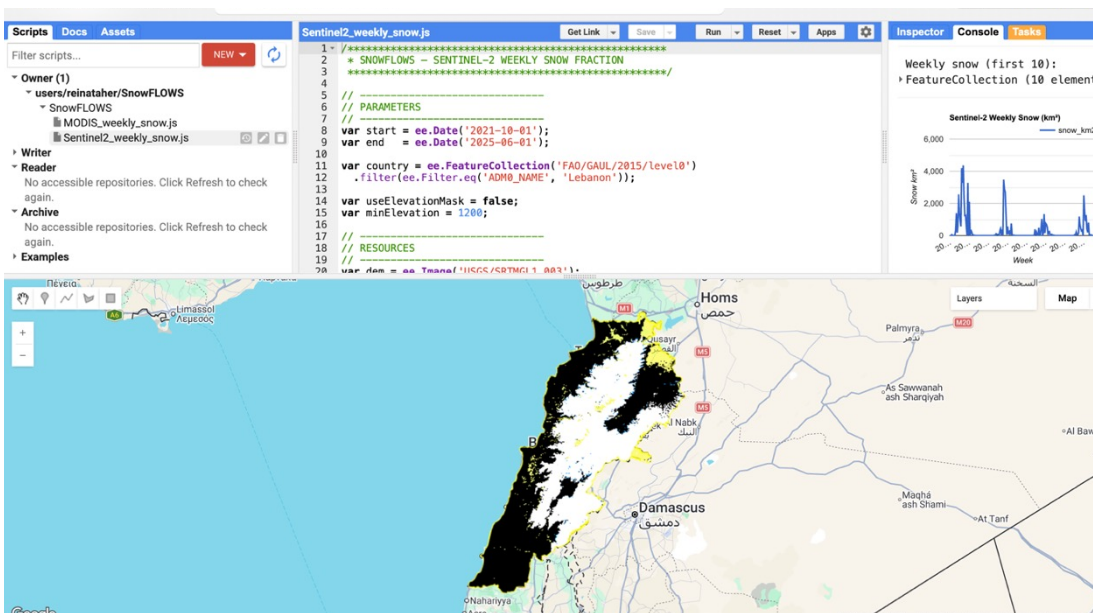
Satellite data:

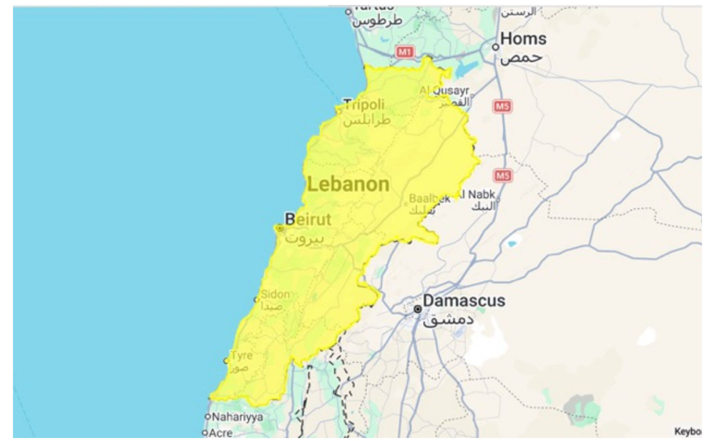
- Sentinel–2 (10m, high spatial resolution)
- MODIS (500m, high temporal frequency)

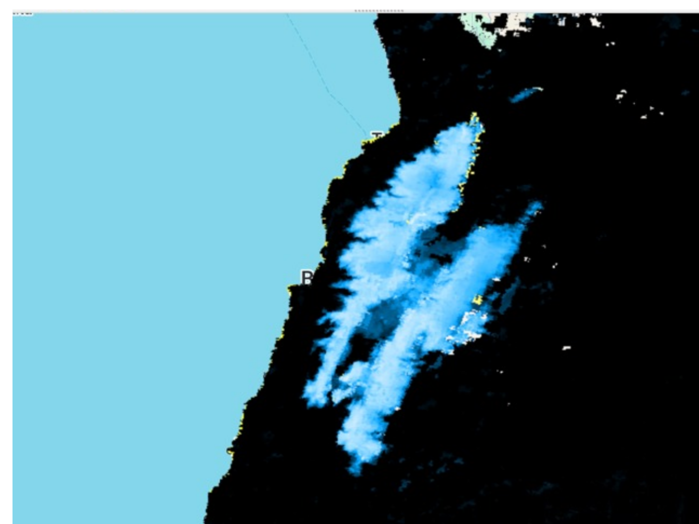
Meteorological data: ERA5 (temperature, precipitation)

Time span: 2016–2025

- – Cloud Masking
- – NDSI Threshold = 0.4

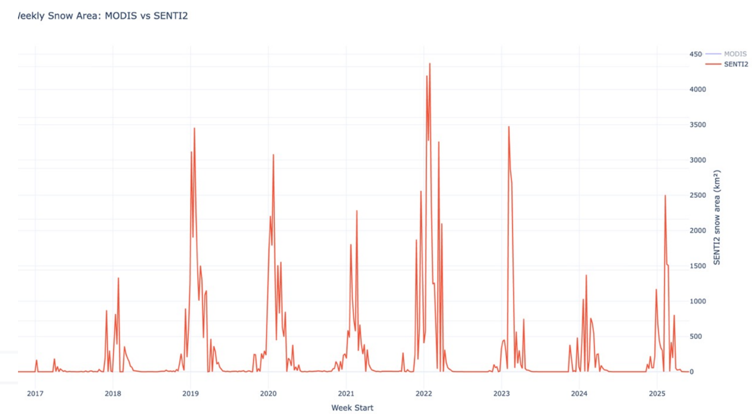


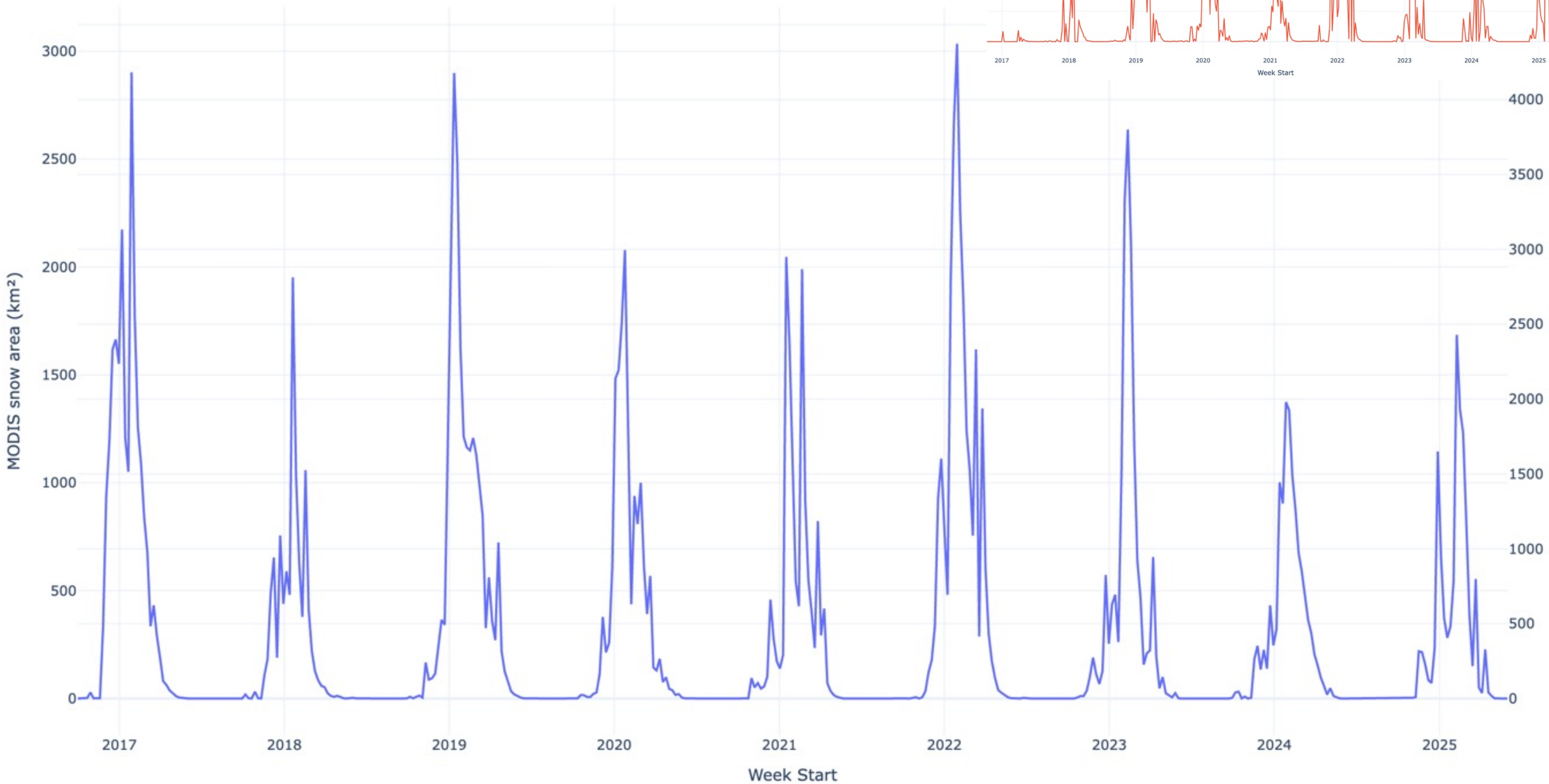




Time Series (MODIS/Senti–2)

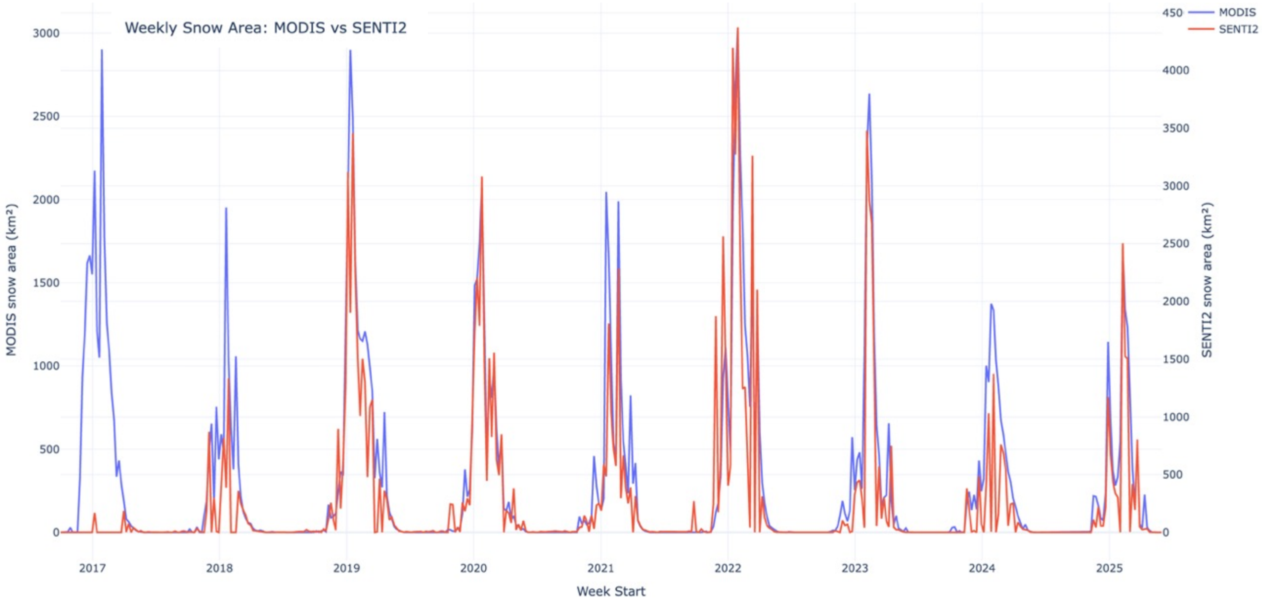
- Weekly aggregation
- 2016–2025
- Strong seasonal pattern

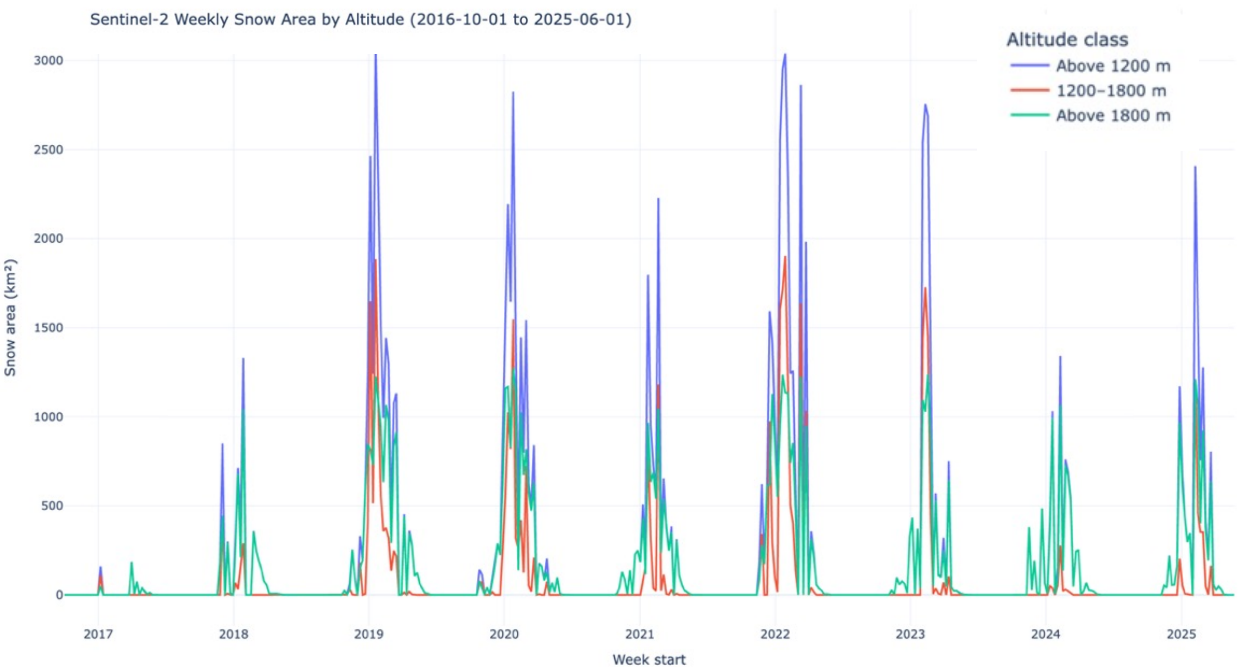


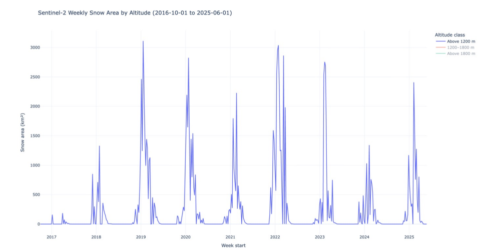


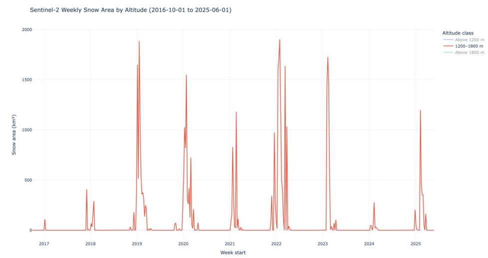
Data Understanding

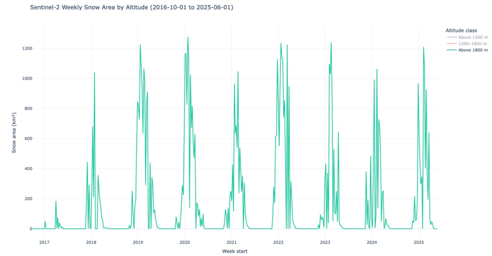
- MODIS = smoother (time)
- Sentinel = detailed (space)
- Higher altitude = longer snow persistence











Dataset + Features + Normalization

year	week_of_year	split	s2_snow_ab	s2_snow_12	s2_snow_ab	modis_snow	modis_snow	modis_snow	temp_1200_	temp_1200_	temp_1800_	precip_1200	precip_1200	precip_1800	target_snow_next_week
2016	39	train	-0.485216	-0.3484557	-0.5684506	-0.3166124	-0.3791174	-0.6657947	0.75902571	0.70377424	0.67435535	-0.6912347	-0.7090942	-0.7409183	-0.485215972
2016	40	train	-0.485216	-0.3484557	-0.5684506	-0.316468	-0.3791174	-0.6657947	0.64039597	0.59769652	0.56474406	-0.6776493	-0.6854045	-0.7155035	-0.485215972
2016	41	train	-0.485216	-0.3484557	-0.5684506	-0.3166124	-0.3791174	-0.6657947	0.39637968	0.35099791	0.33234733	-0.6595996	-0.6823134	-0.6911394	-0.485215972
2016	42	train	-0.485216	-0.3484557	-0.5684506	-0.3127738	-0.3711117	-0.6643295	0.31738927	0.24751189	0.19822102	-0.2261808	0.11516116	0.30817393	-0.485215972
2016	43	train	-0.485216	-0.3484557	-0.5684506	-0.1510835	-0.3430835	-0.6483461	-0.1021078	-0.198233	-0.2281668	0.78999121	0.60147851	0.36653034	-0.485215972
2016	44	train	-0.485216	-0.3484557	-0.5684506	-0.3094166	-0.3776884	-0.6656338	-0.1588268	-0.1778124	-0.1524414	-0.6671165	-0.6728355	-0.7108194	-0.485215972
2016	45	train	-0.485216	-0.3484557	-0.5684506	-0.3053253	-0.3783009	-0.6650553	-0.1530117	-0.1262096	-0.084314	-0.5666062	-0.6077492	-0.6641115	-0.485215972
2016	46	train	-0.485216	-0.3484557	-0.5684506	-0.3051448	-0.3787372	-0.6657947	-0.6398602	-0.725867	-0.7013616	-0.6969052	-0.7105617	-0.7457149	-0.485215972
2016	47	train	-0.485216	-0.3484557	-0.5684506	1.49263016	0.00889353	-0.388541	-0.9393613	-0.9948365	-0.9813	2.81729415	2.45745627	2.14760397	-0.485215972
2016	48	train	-0.485216	-0.3484557	-0.5684506	-0.2461579	0.0242966	2.18162353	-1.1520356	-1.1951726	-1.1912867	-0.2919906	-0.1697063	-0.0924079	-0.485215972

Feature Category	Variables
Seasonal encoding	week_sin, week_cos
Persistence	snow_last_week, snow_last_2weeks
Change dynamics	snow_change_ratio
Aggregated snow signal	snow_mean
Temporal derivatives	snow_gradient, snow_acceleration
Sentinel-2 snow coverage	3 altitude bands
MODIS snow coverage	3 altitude bands
Temperature	3 altitude bands
Precipitation	3 altitude bands

The transformation standardizes each feature according to: $Z = \frac{X-\mu}{\sigma}$

where:

- X is the original feature value
- μ is the feature mean
- σ is the feature standard deviation

From Data → Model

[week1, week2, week3, ..., week8] ⇨ Predict week9

Mathematically:

$X_t = [\, week_{t-7}, \ldots, week_t \,]$

$y_t = snow_{t+1}$

The dataset was therefore reshaped into **3D tensors**:

- (samples, time steps, features)
- Time Steps = 8 weeks
- Features = 20

```
model = Sequential([
    Input(shape=(X_train.shape[1], X_train.shape[2])),

    LSTM(96, return_sequences=True),
    Dropout(0.15),

    LSTM(64),
    Dropout(0.15),

    Dense(32, activation='relu'),
    Dense(16, activation='relu'),

    Dense(1)
])

model.compile(
    optimizer=Adam(learning_rate=0.0003),
    loss=peak_weighted_huber,
    metrics=['mae', rmse]
)

model.summary()
```

Dataset Splitting

Dataset	Percentage
Training set	70%
Validation set	15%
Test set	15%

```
(esibml) reinataher@Reinas-MacBook-Air FYP_1 % python create_sequences.py
Sequences saved successfully
Train shape: (307, 8, 20)
Validation shape: (68, 8, 20)
Test shape: (67, 8, 20)
Number of features used: 20
(esibml) reinataher@Reinas-MacBook-Air FYP_1 %
```

- **Baseline**: $snow_{t+1} \approx snow_t$

```
(esibml) reinataher@Reinas-MacBook-Air FYP_1 % python train_lstm.py
Data loaded successfully
Train: (307, 8, 20)
Validation: (68, 8, 20)
Test: (67, 8, 20)
Baseline persistence MAE: 0.4887628299104479
Model: "sequential"
```

LSTM Model & Loss Function Design

```python

def peak_weighted_huber(y_true, y_pred):
    error = y_true - y_pred
    abs_error = tf.abs(error)

    delta = 1.0

    huber = tf.where(
        abs_error < delta,
        0.5 * tf.square(error),
        delta * (abs_error - 0.5 * delta)
    )

    peak_weight = 1.0 + 1.5 * tf.nn.relu(y_true)

    return tf.reduce_mean(huber * peak_weight)

```

Custom Loss Function

- Peak-weighted Huber Loss
- Combines MAE (robust) + MSE (sensitive)
- Higher weight on extreme snow values

→ **Improves learning of peak snow events**

```python
model = Sequential([
    Input(shape=(X_train.shape[1], X_train.shape[2])),

    LSTM(96, return_sequences=True),
    Dropout(0.15),

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])

model.compile(
    optimizer=Adam(learning_rate=0.0003),
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)

model.summary()
```
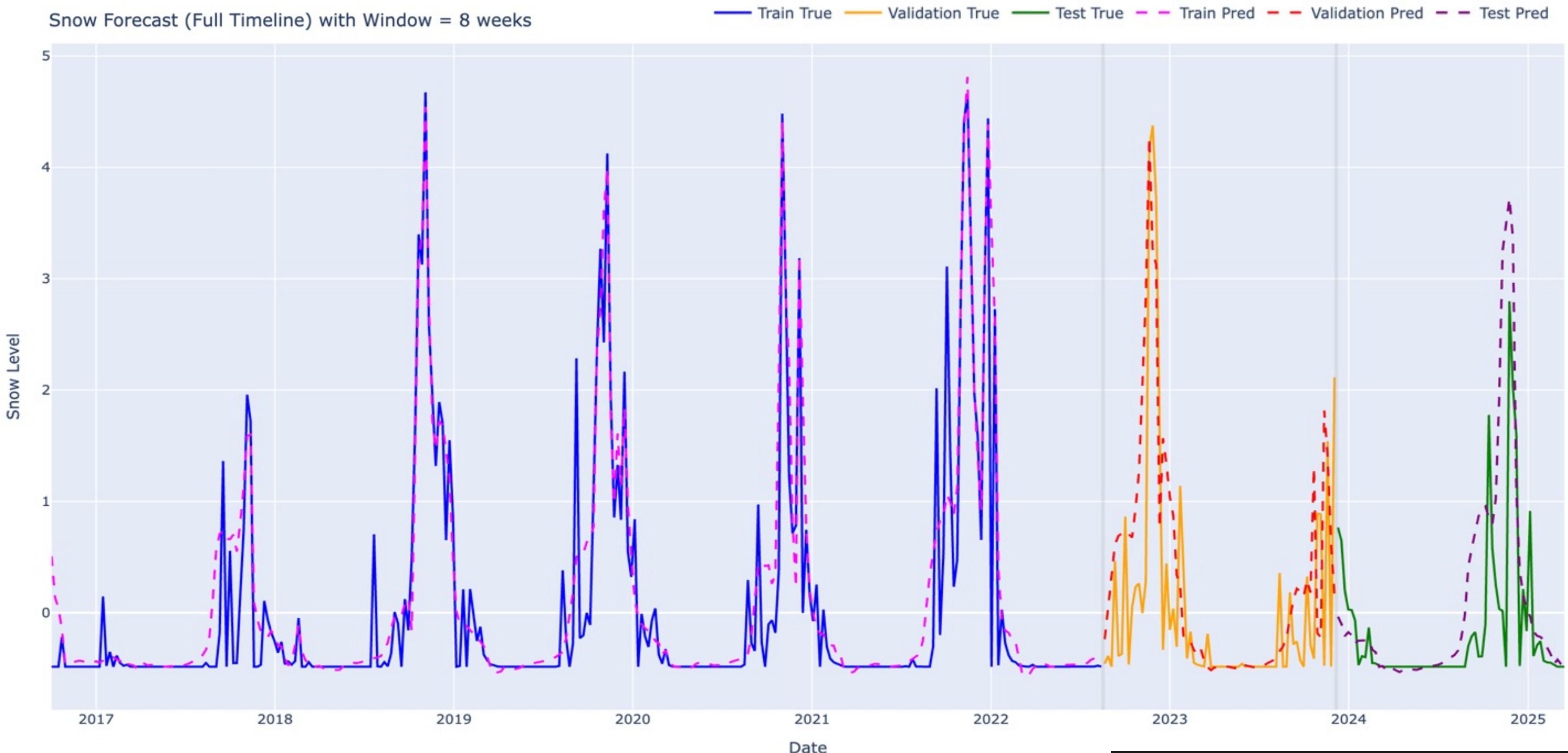

Why LSTM ?

- Captures temporal dependencies
- Handles sequential environmental data
- Learns both short-term & long-term patterns

Best Model Example

For a sequence length of 8–week:

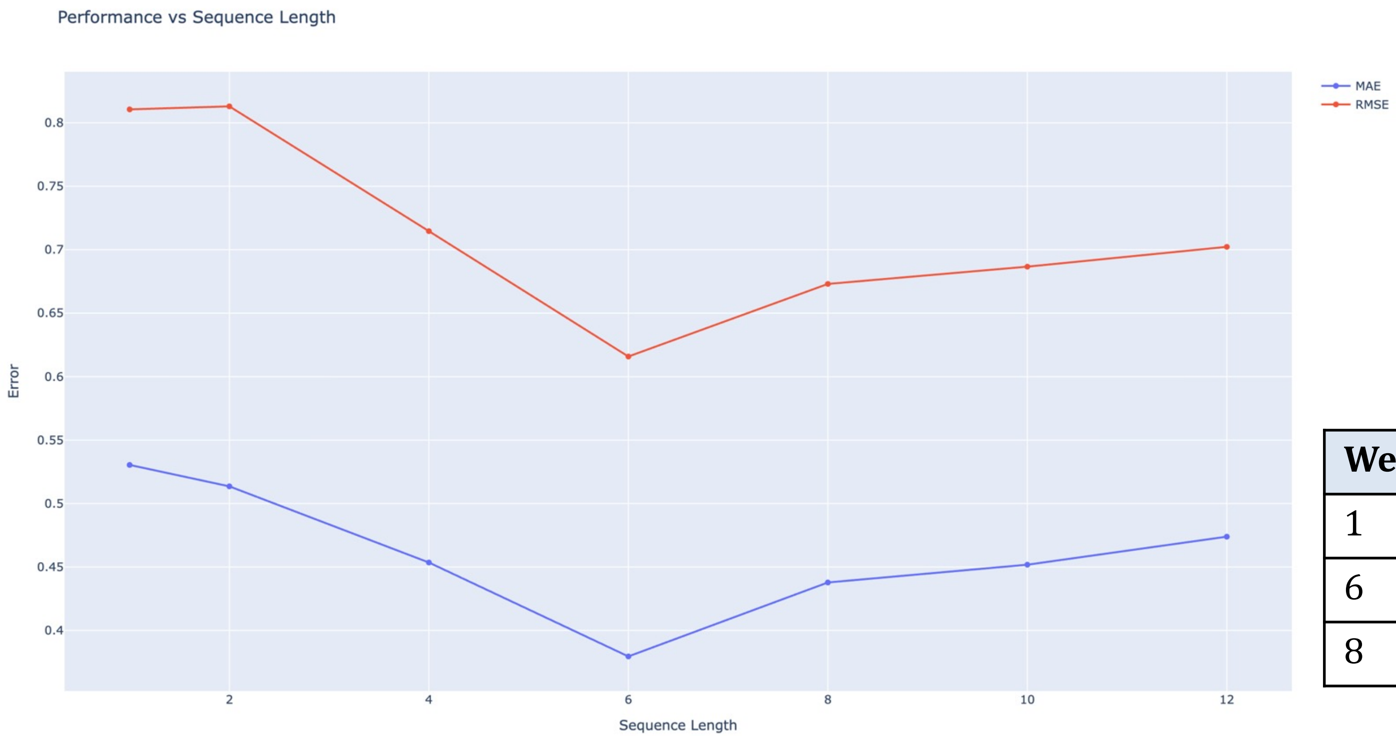


Key Observations

- LSTM significantly outperforms the baseline model
- Improvement is especially clear in RMSE reduction
- Model generalizes well on unseen test data
- Errors increase slightly on test set due to environmental variability

```
MODEL PERFORMANCE (TEST SET)
Test MAE: 0.4147394001483917
Test RMSE: 0.6626249551773071

MODEL PERFORMANCE (TRAIN SET)
Train MAE: 0.21862621234184623
Train RMSE: 0.4474281788195303

BASELINE PERFORMANCE (PERSISTENCE MODEL)
Baseline MAE: 0.4887628299104479
Baseline RMSE: 0.8992733218617118
```

Results of Sequence Length Experiments & Trade–Off

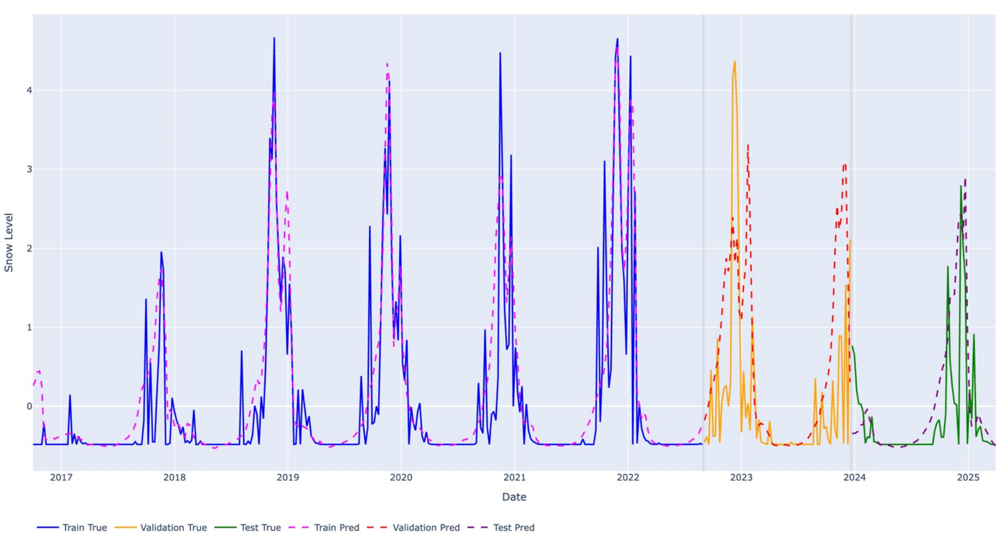


Weeks	MAE
1	high
6	**best**
8	slightly worse

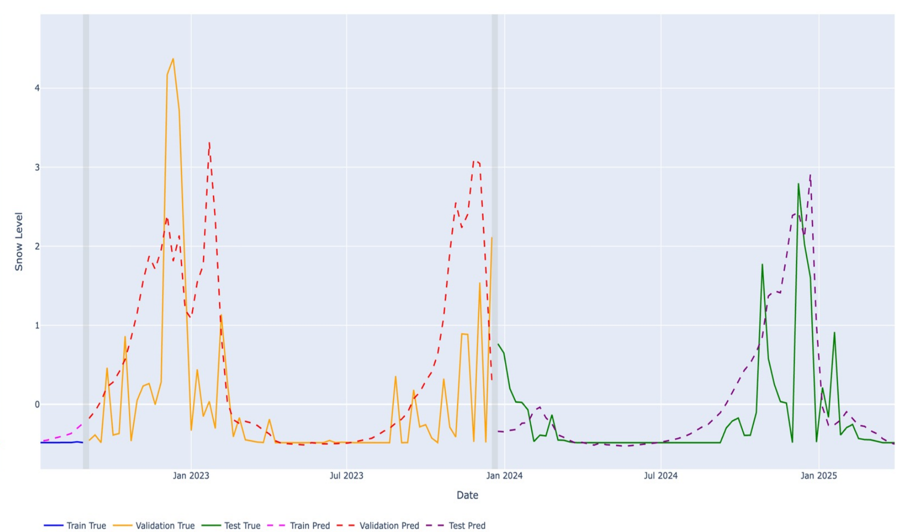
Sequence (weeks)	MAE (mean)	MAE (std)	RMSE (mean)	RMSE (std)
1	0.530	0.011	0.811	0.010
2	0.514	0.014	0.813	0.014
4	0.454	0.018	0.715	0.018
6	**0.380**	**0.012**	**0.616**	**0.015**
8	0.438	0.046	0.673	0.047
10	0.452	0.046	0.687	0.048
12	0.474	0.057	0.702	0.058

Results of Sequence Length Experiments & Trade–Off

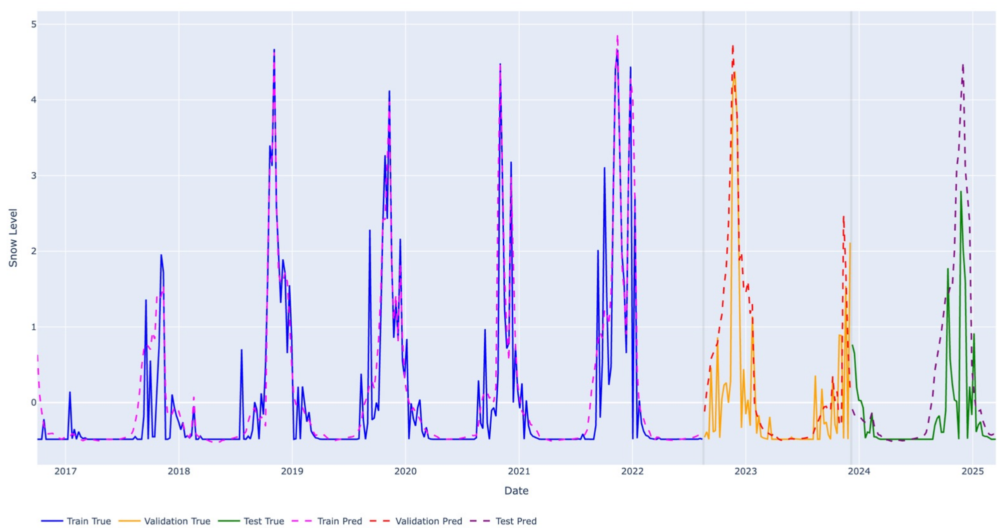
Snow Forecast (Full Timeline) with Window = 6 weeks



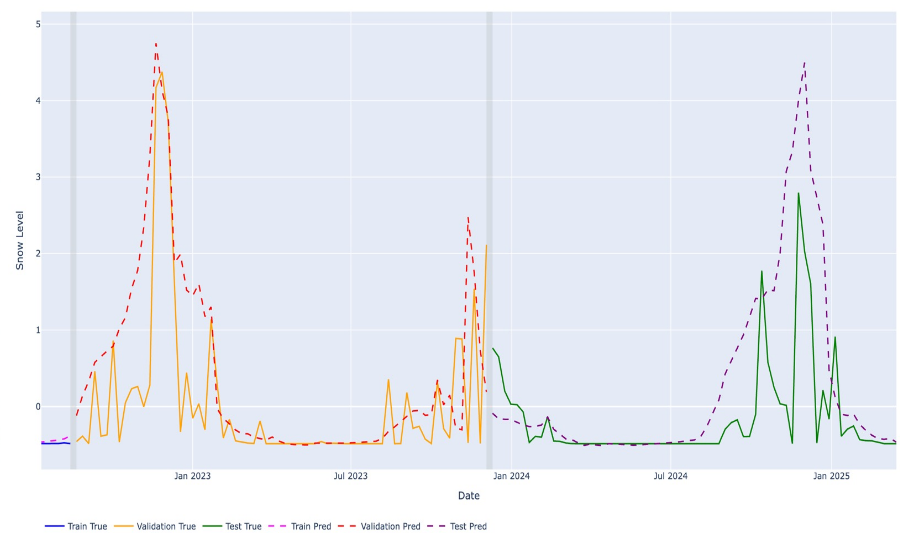
Snow Forecast (Full Timeline) with Window = 6 weeks



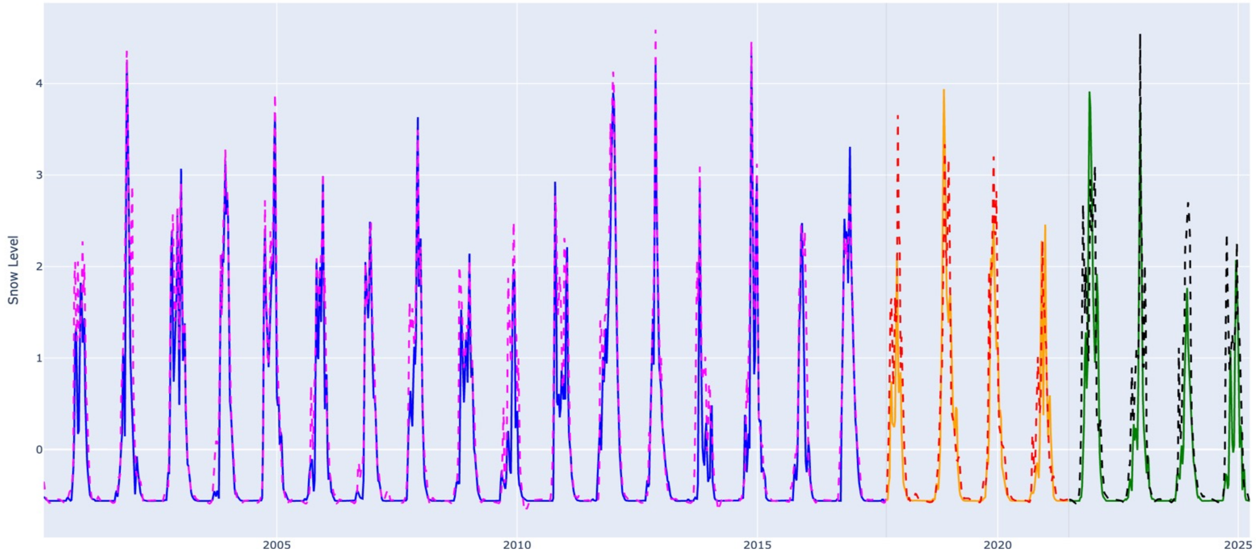
Snow Forecast (Full Timeline) with Window = 8 weeks

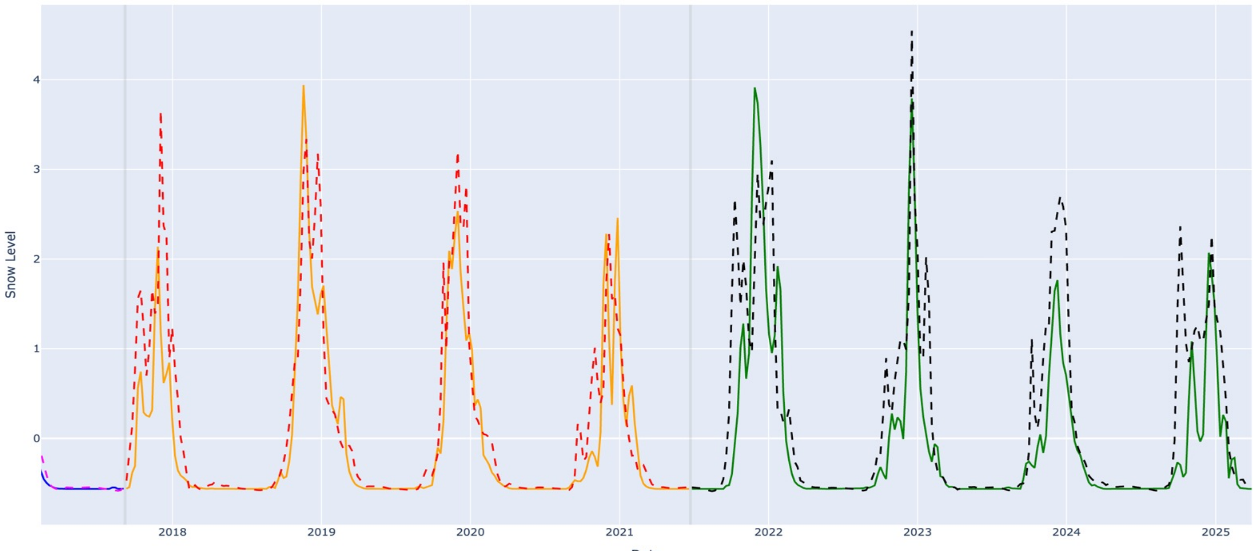


Snow Forecast (Full Timeline) with Window = 8 weeks



Long–Term Forecasting Experiment (MODIS 25–Years Dataset)





Objective

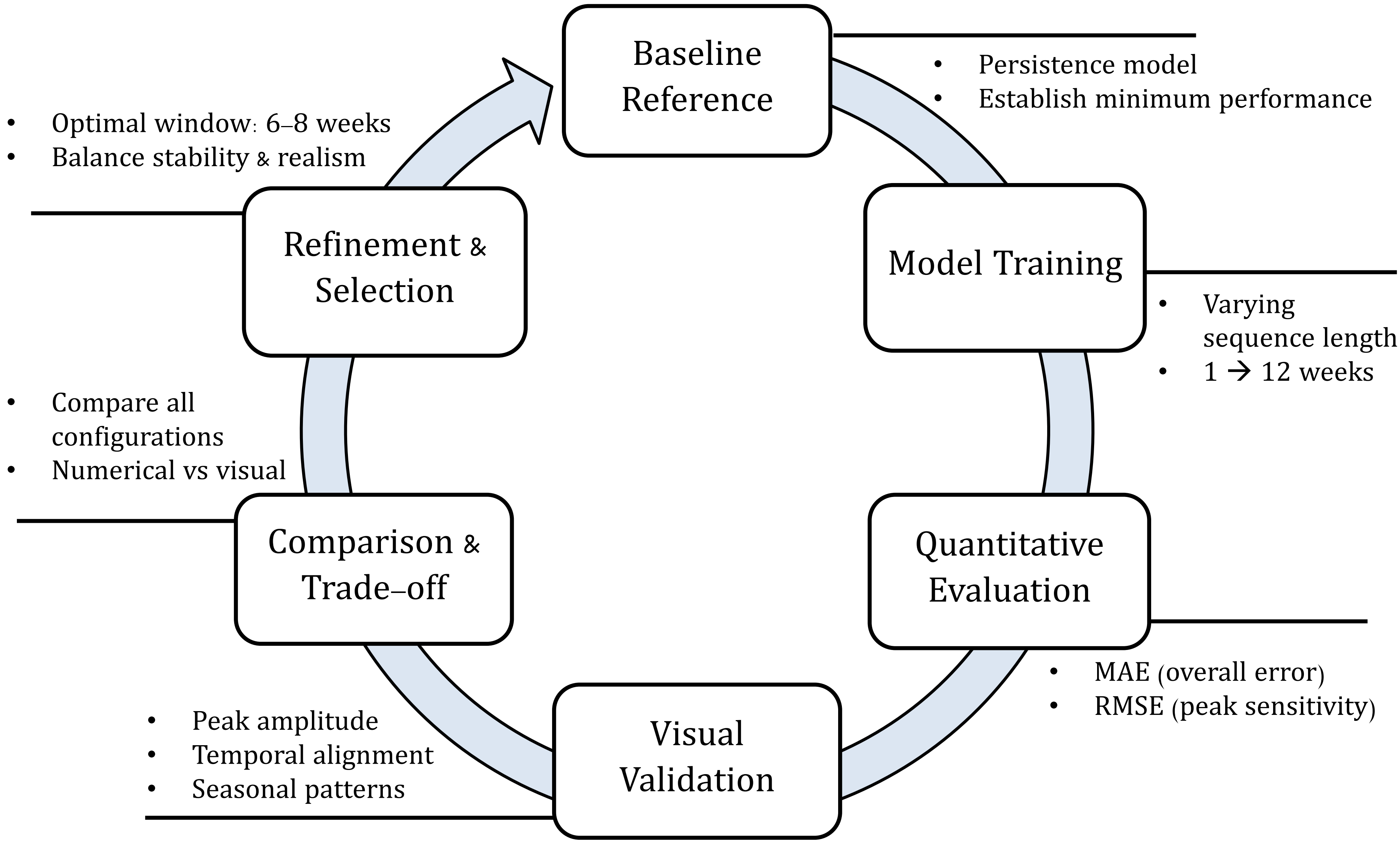
- Test whether more historical data improves performance

Observations

- Smoother long–term trends
- Better seasonal consistency
- Still noticeable prediction errors (RMSE ≈ 0.6)

→ *More data improves stability, but does NOT eliminate forecasting errors*

Testing, Evaluation and Iterative improvements



Solution Impact

Environmental Impact
- Improved monitoring of snow cover dynamics
- Better understanding of water resources availability
- Supports climate change analysis in Lebanon

Societal Impact
- Better preparedness for snowfall events
- Supports road safety & accessibility
- Helps authorities in decision–making

Economic Impact
- Supports winter tourism (ski resorts)
- Helps agriculture planning (water from snowmelt)
- Reduces uncertainty in resource management

Limitations & Future Work

Limitations

- Trade–off between sequence lengths (6 vs 8 weeks)
- Limited temporal coverage of Sentinel–2 data
- Difficulty capturing extreme snow peaks
- Evaluation focused on MAE & RMSE
- Project scope constrained by time

Future Work

- Hybrid model combining Sentinel–2 + MODIS
- Advanced architectures (Attention / Transformers)
- Additional evaluation metrics
- Extend dataset
- Integrate physical constraints into ML models

Conclusion

Key Achievements

- Built a complete snow forecasting pipeline
- Integrated multi–source data (Sentinel–2, MODIS, ERA5)
- Developed and optimized an LSTM model
- Outperformed baseline methods

Main Findings

- Snow dynamics are complex and nonlinear
- Trade–off between accuracy and realism
- Performance depends on model design, not only data quantity